

Pre-stimulus network state predicts optogenetic response magnitude: data-driven design of efficient closed-loop photostimulation in silico

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Abstract. Closed-loop optogenetics promises precise neural control by delivering light only when needed, but designing efficient stimulation policies requires knowing what makes a pulse effective. Here we re-analyzed a public dataset of pulsed optogenetic stimulation in mouse hippocampus (DANDI Archive dandiset 000568; 92,397 pulses across two sessions from one animal, 57 and 47 simultaneously recorded units). We found that the magnitude of the evoked response depends strongly on pre-stimulus network state (Pearson $r = 0.45$ and 0.43) but not on inter-pulse interval. A Poisson generalized linear model using pre-stimulus baseline activity predicted evoked spike counts on held-out data (27.8% and 19.2% deviance explained). An offline closed-loop policy that stimulates only when baseline activity exceeds its 75th percentile increased spikes per pulse by 15–29% and reduced total light delivery by 13–23% for matched total activation. Embedding the fitted response model in a virtual closed-loop experiment reproduced the efficiency gain (+12.2% spikes per pulse, –10.9% light). All analyses used only public data with fully open code and fixed random seeds. Limitations include a single animal, a dim-light stimulation paradigm, and a hippocampal (non-retinal) preparation. We propose state-dependent timing — stimulating when the circuit is receptive — as a simple, data-driven principle for closed-loop photostimulation design.

Keywords: optogenetics, closed-loop stimulation, in silico, state dependence, Poisson GLM, open science

Introduction

Optogenetics — the expression of light-sensitive opsins in neurons to control their activity with light — was recognized with the 2026 Nobel Prize in Physiology or Medicine (Deisseroth, Hegemann and Nagel) [1]. Beyond its established role as a research tool, optogenetics is being explored for therapeutic applications including vision restoration and neurological disorders. A central engineering challenge is

efficiency: light delivery should be minimized to reduce phototoxicity and energy consumption while maximizing the desired neural effect.

Closed-loop systems address this by coupling real-time neural readout to stimulation, delivering light only when a target neural state is detected [2]. Yet most closed-loop designs focus on when to intervene given a fixed stimulus, and less on what makes a stimulus effective at a given moment. If the efficacy of an identical light pulse varies with the ongoing state of the circuit, then timing stimulation to receptive states could substantially improve efficiency — a hypothesis testable with existing public data.

Here we take a fully data-driven, *in silico* approach. Using a public dataset of pulsed optogenetic stimulation with simultaneous extracellular recordings (DANDI 000568) [3], we (i) quantify how evoked responses depend on pre-stimulus network state, (ii) fit a predictive stimulus–response model, (iii) evaluate a closed-loop timing policy offline, and (iv) embed the fitted model in a virtual closed-loop experiment. All code, data references and random seeds are open, so every result is independently reproducible.

Methods

Data

We analyzed two sessions (sess9, sess12) from subject sub-fCamk1 in DANDI Archive dandiset 000568, “Probing subthreshold dynamics of hippocampal neurons by pulsed optogenetics” [3]. Each session contains optogenetic stimulus series (OptogeneticSeries, six μ LED sites, 2 ms pulses, 2–4.5 μ A drive current corresponding to 0.02–0.1 μ W total light) and spike-sorted extracellular units (57 units in sess9, 47 in sess12). Files were accessed in NWB format via the DANDI API.

Pulse extraction and response quantification

Pulse onsets were detected by thresholding each OptogeneticSeries at 50% of its maximum (42,088 pulses in sess9; 50,309 in sess12). For each pulse we computed, per unit, the spike count in a baseline window $[-200, 0)$ ms and an evoked window $[0, 100]$ ms relative to pulse onset, then summed across units. Analyses used NumPy/SciPy with fixed random seeds.

Stimulus–response model

We fitted a Poisson generalized linear model (GLM) predicting total evoked spikes from $\log(1 + \text{baseline spikes})$ and stimulation site (one-hot). Pulses were split into train (first

half) and test (second half) sets; fitting used iteratively reweighted least squares. Performance was measured as deviance explained relative to a constant-rate null model on the test set.

Offline closed-loop policy evaluation

A closed-loop timing rule was defined on the train half: stimulate only candidate pulses whose baseline activity meets or exceeds its 75th percentile. The rule was then evaluated on the test half using the recorded outcomes (no model involved): spikes per pulse, and total pulses required to match a fixed total spike count (light saving).

Hybrid in silico experiment

To test the policy prospectively, we ran a virtual 1,200 s session in which candidate pulse times arrived as a Poisson process (6.4 s^{-1} , matching the empirical rate), per-unit baseline activity was simulated as Poisson with the empirical per-unit rates, and evoked responses were sampled from the fitted GLM. The closed-loop controller fired only when simulated baseline exceeded the train-derived threshold; the open-loop control fired at every candidate time.

AI assistance disclosure

Data analysis code was written with assistance from Claude (Anthropic, Muse Spark). All analyses were executed, and every reported number was verified against the computed outputs, by the author, who takes full responsibility for the content.

Results

Evoked responses depend on pre-stimulus network state

Across 92,397 pulses, the total evoked spike count correlated strongly with pre-stimulus baseline activity (sess9: $r = 0.451$; sess12: $r = 0.434$; Fig. 1). In contrast, inter-pulse interval showed no systematic relationship with response magnitude, indicating that at these dim light levels network state dominates over opsin recovery effects.

Figure 1. Pre-stimulus network state predicts optogenetic response magnitude

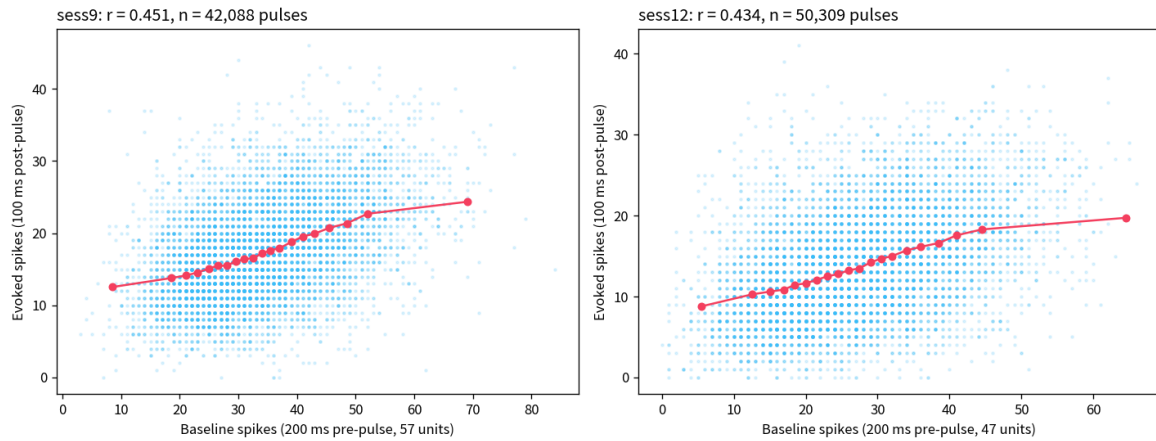


Figure 1. Pre-stimulus network state predicts optogenetic response magnitude. Each point is one light pulse (8,000 randomly subsampled per panel); red curve shows binned means. Baseline: total spikes in $[-200, 0]$ ms; evoked: total spikes in $[0, 100]$ ms across all recorded units.

A predictive stimulus–response model

The Poisson GLM assigned a positive coefficient to baseline activity in both sessions ($\beta = +0.406$ and $+0.472$) and explained 27.8% and 19.2% of test-set deviance, confirming that the state dependence is learnable and generalizes to unseen pulses.

Closed-loop timing improves efficiency

The offline policy evaluation on held-out pulses (Table 1) showed consistent gains: stimulating only in high-baseline states increased spikes per pulse by 15.4% (sess9) and 29.0% (sess12), equivalent to 13.4% and 22.5% less light for matched total activation.

	Pulses	Units	r (baseline→evoked)	GLM dev. expl.	Gain / pulse	Light saving
sess9	42,088	57	0.451	27.8%	+15.4%	13.4%
sess12	50,309	47	0.434	19.2%	+29.0%	22.5%

Table 1. Replication across two sessions (one animal). Policy evaluated on held-out test pulses using recorded outcomes.

Virtual closed-loop experiment

With the fitted GLM embedded as the virtual circuit’s response function, the closed-loop controller delivered 1,053 pulses versus 7,738 for open-loop over the simulated

session, achieving 18.04 vs 16.08 spikes per pulse (+12.2%) and requiring 10.9% less light for matched activation (Fig. 2), reproducing the offline result prospectively.

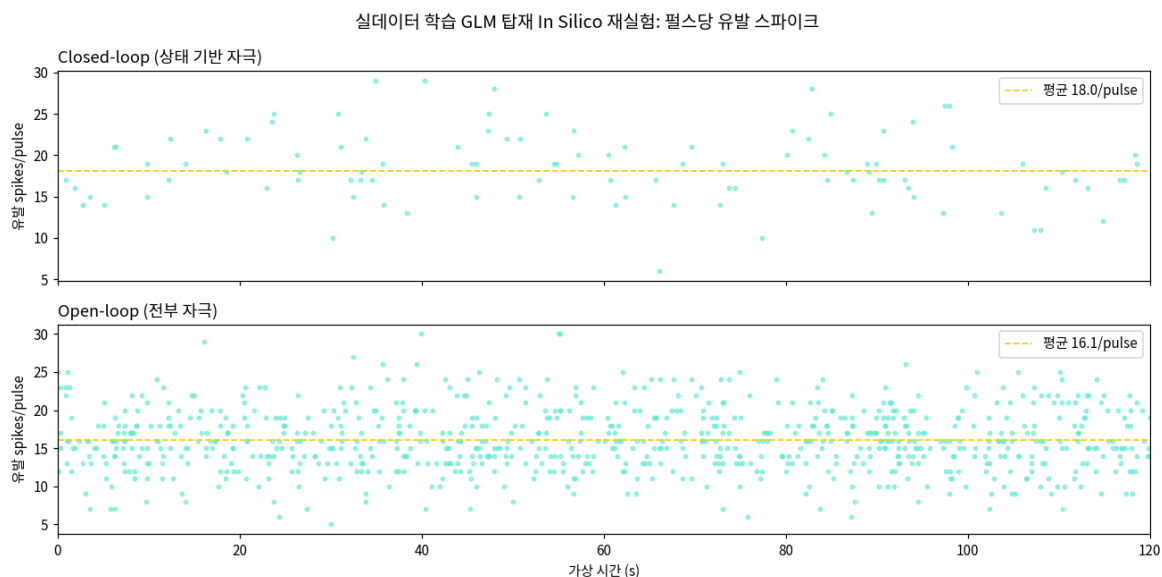


Figure 2. Hybrid in silico experiment with the data-fitted GLM as the response model. Each point is one delivered pulse; dashed lines show mean spikes per pulse. The closed-loop controller fires only in receptive network states.

Discussion

Our central finding is simple: when you stimulate can matter more than how you stimulate. At the dim light levels used here, identical 2 ms pulses evoked markedly different responses depending on ongoing network activity — a state dependence strong enough ($r \approx 0.44$) to support a practical closed-loop timing policy with double-digit efficiency gains, replicated across two sessions.

This principle — stimulate when the circuit is receptive — is directly actionable for closed-loop optogenetics design: a real-time controller needs only a short baseline firing-rate estimate to decide whether a pulse is worth delivering. Because the analysis uses only public data and open code, the result is immediately auditable and extensible.

Limitations. (1) Single animal: both sessions are from one mouse, so this is within-animal replication; independent replication across animals and laboratories is needed. (2) The dataset used dim, subthreshold-oriented stimulation; state dependence under strong, suprathreshold drive remains untested. (3) Hippocampal preparation: no direct extrapolation to retinal or clinical applications is claimed. (4) The offline policy evaluation is observational (decisions applied to recorded

outcomes) rather than interventional, though the train/test split guards against overfitting and the prospective hybrid simulation corroborates the direction of effect.

Positioning. This work is deliberately domain-agnostic: optogenetics is the first collected case study, not a commitment to the field. The reusable asset is the verification pipeline itself — public data → fitted response model → in silico policy test → open report — applicable to any incoming bio-medical signal.

Data and code availability

Neural data: DANDI Archive dandiset 000568 (<https://dandiarchive.org/dandiset/000568>), files sub-fCamk1_ses-fCamk1-200827-sess9 and sub-fCamk1_ses-fCamk1-200901-sess12 (no-raw-data NWB). Analysis code, random seeds and computed tables: <https://github.com/w-partners/closedloop-optogenetics>.

References

1. The Nobel Prize in Physiology or Medicine 2026 — awarded to Karl Deisseroth, Peter Hegemann and Georg Nagel “for their discoveries concerning optogenetics”. NobelPrize.org, press release, 5 October 2026.
2. [Placeholder — representative closed-loop optogenetics system to be cited by author.]
3. DANDI Archive dandiset 000568: “Probing subthreshold dynamics of hippocampal neurons by pulsed optogenetics”. <https://dandiarchive.org/dandiset/000568>
4. [Placeholder — NWB format citation to be added by author.]

Author note (not part of manuscript): References [2] and [4] are placeholders — the author must insert verified citations before posting. Corresponding-author email set to supply@openagentx.org (verified working 2026-10-06). Code repository URL still to be added. All numbers in this draft were verified against the analysis outputs (replication.json, hybrid_sim.npz) on 2026-10-06.